Natural Language Processing
DSL504/DS504
Mid-Semester Examination
Time 2 hours

1. Find the edit distance between "hotel" and "Chore" using the dynamic programming approach. Show the matrix of distances among prefixes.

[5 Marks]

2. Consider the following paragraph:

"Marie was born in 1867 in Warsaw, Poland, where her father was a professor of physics. At an early age, she displayed a brilliant mind and a blithe personality. Her great exuberance for learning prompted her to continue with her studies after high school. She became disgruntled, however, when she learned that the university in Warsaw was closed to women. Determined to receive a higher education, she defiantly left Poland and in 1891 entered the Sorbonne, a French university, where she earned her master's degree and doctorate in physics."

Create a bi-gram language model from the paragraph without smoothing. Detect whether there is any real-word or non-word error in the following phrase - "She learned hert". Assume the vocabulary to be only the words contained in the paragraph.

[5 Marks]

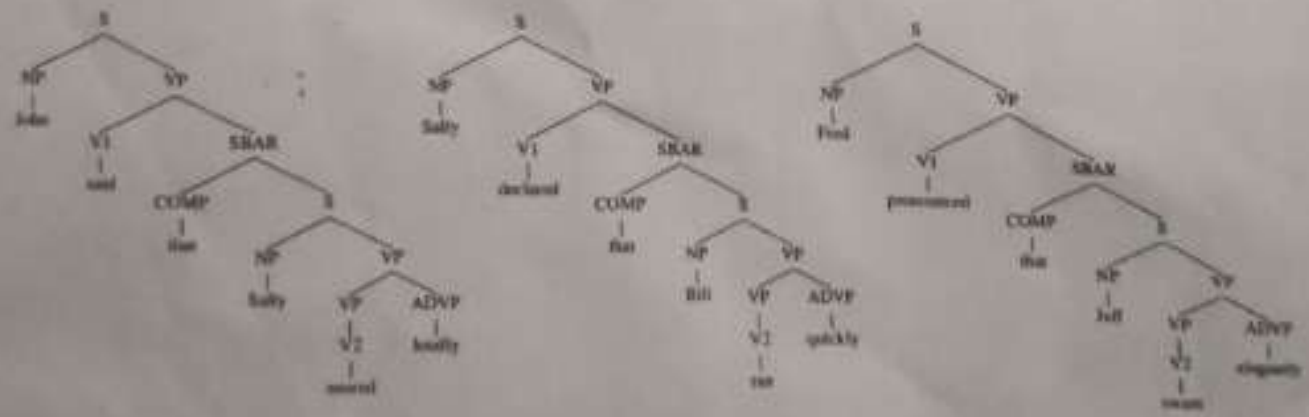
3. Consider the following CFG:

S → NP VP	PP → Pre NP
S → *I* VP PP	V → *ate*
NP → Det N	Det → *the* | *a*
VP → *ate* NP	N → *fork* | *salad*
VP → V	Pre → *with*

Convert it to Chomsky Normal Form.

[3 Marks]

4. Suppose, you obtain the following three dependency parsing trees from a tree bank.



Show the PCFG that can be derived from these three trees.

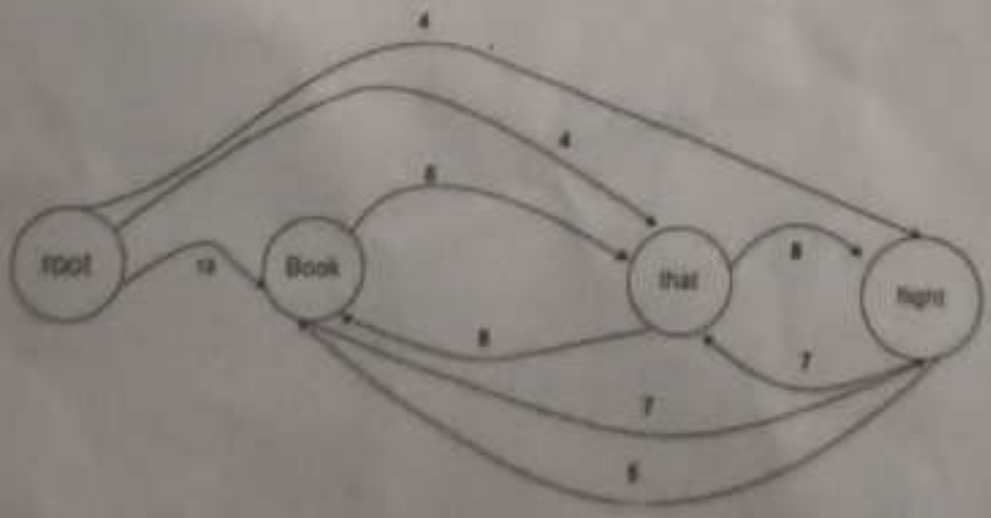
[4 Marks]

3. Given the PCFG in the table below, use CKY parser to parse the sentence "A student likes writing tests". Determine probabilities of all the generated parse trees obtained (there can be more than one).

Parent	Child / Children	Probability
S	NP VP	1
VP	VBG NNS	0.3
VP	VBZ VP	0.4
VP	VBZ NP	0.3
NP	DT NN	0.5
NP	JJ NNS	0.5
DT	"A"	1
NN	"student"	1
VBZ	"likes"	1
VBG	"writing"	1
JJ	"writing"	1
NNS	"tests"	1

[5 Marks]

4. Suppose, we wish to apply MST based parsing for the sentence - "Book that flight". Apply the "Chu-Liu-Edmonds" algorithm to the following graph to obtain the MST which gives us the dependency relations.



[6 marks]

7. Suppose you want to use a bigram POS tagger to tag the phrase, "the light book", where we have the following probabilities:

P(Verb) = 0.2, P(Noun) = 0.3, P(Det) = 0.05, P(Adj) = 0.1
P(the|Det) = 0.3, P(light|Noun) = 0.003, P(light|Adj) = 0.002, P(light|Verb) = 0.06, P(book|Noun) = 0.003, P(book|Verb) = 0.01
P(Verb|Det) = 0.00001, P(Noun|Det) = 0.5, P(Adj|Det) = 0.3, P(Noun|Noun) =0.2, P(Noun|Adj) = 0.2, P(Noun|Verb) = 0.3, P(Verb|Noun) = 0.3, P(Verb|Adj) = 0.001, P(Verb|Verb) = 0.1

Work out in detail the steps of the Viterbi algorithm. You can use a Table to show the steps. Assume all other conditional probabilities, not mentioned to be zero. [7 marks]

8. a. *What is negative sampling* and what is its benefit? Explain with an example using the objective *function for a standard word2vec* model.

b. Suppose, you got a trained word2vec model where the following word vectors are given: drive - [0.3, *0.4], driver [0.8, 0.7], swim [0.4,* 0.1]. From this can you calculate the word vector for the word *"swimmer"?*

[3 + 2 Marks]